14.2 Phase Changes

Question Paper

Course	CIE A Level Physics
Section	14. Temperature
Topic	14.2 Phase Changes
Difficulty	Hard

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

Four identical ice spheres are dropped into a thermally isolated cylinder containing water.

The specific heat capacity of the ice is 1.9 kJ kg^{-1} K^{-1} and the specific heat capacity of water is 5.7 kJ kg^{-1} K^{-1}.

Sketch a graph on Fig. 1.1 to show the changes in temperature of the molecules within the ice cubes with time, from the point they are added to the water until they are in thermal equilibrium with the water molecules.

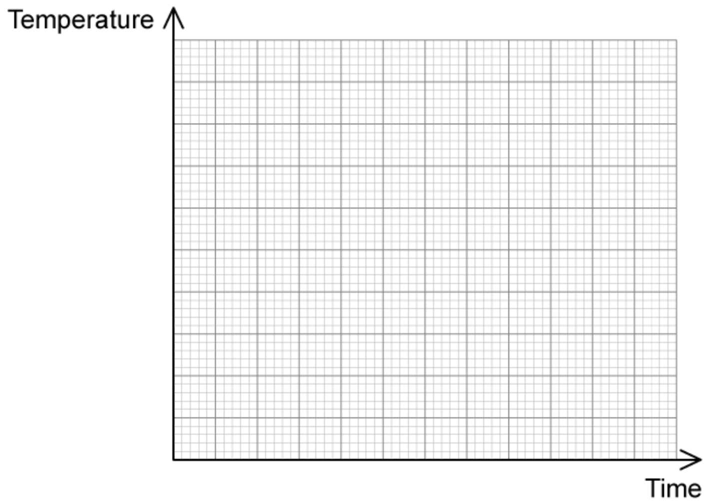


Fig. 1.1

[4 marks]

Question 1b

The ice spheres have a radius of 20 mm and a density of 870 kg m^{-3}. Their initial temperature is –7.2 °C. The specific latent heat of fusion of ice is 442 kJ kg^{-1}. The water has an initial temperature of 22.2 °C and a mass of 0.75 kg.

Determine the final temperature of the water.

[5 marks]

Question 1c

The experiment is repeated with four identical cubes of ice. The cubes have equal volume, mass and density to the spheres from parts **(a)** and **(b)**.

Determine through calculation whether the final temperature would be obtained quicker when using the cubes instead of the spheres.

[5 marks]

Question 2a

As part of a model village pyrotechnics display a 33 g iron ball bearing, cooled to a soecific initial temperature of –2.1592 °C, is dropped into a pool. The ball begins to accelerate uniformly from a height of 0.75 m above the surface of the pool with a velocity of 20 m s^{-1}.

Assume that the ball falls with a constant acceleration due to free fall but a thermal energy transfer takes place between the ball and the air.

The iron ball has a specific heat capacity of 440 J kg^{-1} °C^{-1}.

Calculate the temperature of the ball immediately before it hits the water.

[4 marks]

Question 2b

The iron ball bearing from **(a)** falls into the pool. The pool has dimensions of 5 cm, 10 cm and 30 cm. The density of the water in the pool is 1030 kg m^{-3} and has a specific heat capacity of 4300 J kg^{-1} °C^{-1}. The initial temperature of the water in the pool is 22.5173 °C.

Calculate the final temperature of the ball and the water after impact.

[3 marks]

Question 3a

A professional cyclist of mass 50 kg is testing a new hydraulic brake system on his racing bike of mass 5 kg. He is travelling at a constant speed of 25 m s^{-1} when he has to brake suddenly.

The four brake discs on the bike each have a mass of 250 g, included in the mass of the bike. The specific heat capacity of the mineral oil in the brake system is 1.77 kJ kg^{-1} °C^{-1}.

Calculate the overall increase in the temperature of the brake discs.

[2 marks]

Question 3b

Explain how the temperature difference in the brake disc fluid can be reduced.

[2 marks]

Question 3c

Water has a specific heat capacity of 4200 J kg^{-1} °C^{-1} and a boiling point of 100 °C. The boiling point of mineral oil is 300 °C.

Explain whether adding water to the mineral oil in the brake fluid will make it safer when it overheats.

[3 marks]

Model Answers: Hard

1a

(a) A graph that would score 4 marks should look like this:

List the known quantities:

- Specific heat capacity of ice = 1.9 kJ kg^{-1} K^{-1}
- Specific heat capacity of water = 5.7 kJ kg^{-1} K^{-1}

First section:

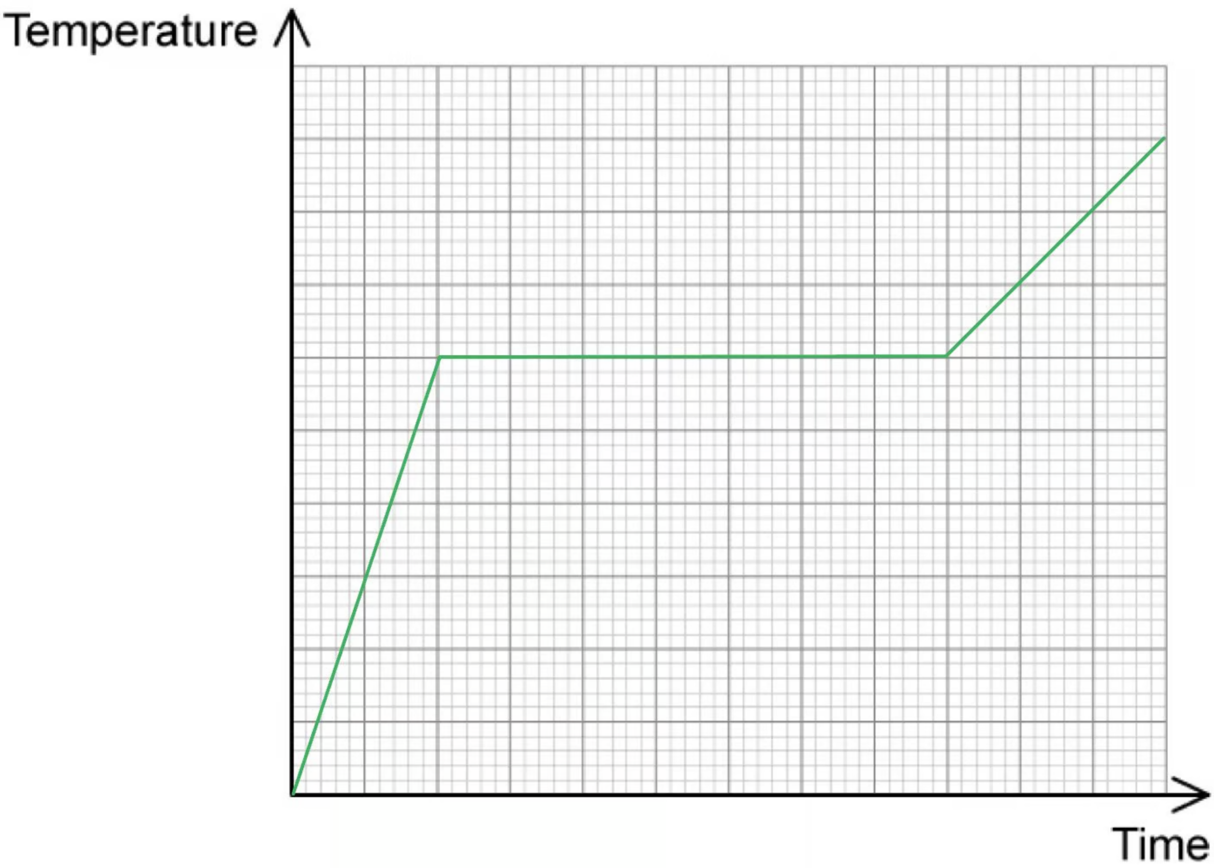
- After the ice is added to the water, being added to the water, the ice will increase its temperature until its melting point
- This is shown by the positive gradient of the diagonal line; [1 mark]

Second section:

- The ice remains at the same temperature whilst melting into a liquid
- This is shown by the horizontal line; [1 mark]

Third section:

- After melting the ice will increase its temperature as a liquid
- The specific heat capacity of the ice is twice that of the water, hence, its temperature will increase twice as quickly
- This is shown by the positive gradient of the diagonal line; [1 mark]
- With a gradient a third of the value of that in the first section; [1 mark]



[Total: 4 marks]

Use the **information provided** on specific heat capacity to plot the graph, it is provided for a reason.

1b

(b) Determine the final temperature of the water:

List the known quantities:

- Specific heat capacity of ice, c_i = 1.9 kJ kg^{-1} K^{-1} = 1.9 × 10^3 J kg^{-1} K^{-1} (from part (a))
- Specific heat capacity of water, c_w = 5.7 kJ kg^{-1} K^{-1} = 5.7 × 10^3 J kg^{-1} K^{-1} (from part (a))
- Radius of ice sphere, r = 20 mm = 20 × 10^{-3} m
- Density of ice, ρ = 870 kg m^{-3}
- Initial temperature of water, T_w = 22.2 °C
- Initial temperature of ice, T_i = –7.2 °C
- Mass of water, m_w = 0.75 kg
- Specific latent heat of fusion, L = 442 kJ kg^{-1} = 442 × 10^3 J kg^{-1}

Calculate the volume of the 4 ice spheres, V:

- Volume of sphere = $\frac{4}{3}\pi r^3$
- Volume of 4 spheres, $V = 4 \times \frac{4}{3}\pi r^3$
- $V = 4 \times \frac{4}{3}\pi(20 \times 10^{-3})^3 = 4 \times (3.35 \times 10^{-5})$
- $V = 1.34 \times 10^{-4}\ \text{m}^3$ [1 mark]

Calculate the mass of the ice spheres, m_i:

- Recall the equation for density: $\rho = \frac{m_i}{V}$ so $m_i = \rho V$
- $m_i = 870 \times (1.34 \times 10^{-4})$
- $m_i = 0.12\ \text{kg}$ [1 mark]

Determine the equation for the energy gained by the ice spheres, Q_i:

- Energy gained by ice = energy required to heat ice + energy required to melt ice + energy required to heat water
- Energy gained by ice, $Q_i = (m_i c_i \Delta T_i) + (m_i L) + (m_i c_w \Delta T_w)$
- $Q_i = (0.12 \times (1.9 \times 10^3) \times (0 - -7.2)) + (0.12 \times (442 \times 10^3)) + (0.12 \times (5.7 \times 10^3)\, T_f)$
- $Q_i = 54681.6 + 684T_f$ [1 mark]

Determine an equation for the energy change in the water, Q_w:

- Energy lost by water, $Q_w = m_w c_w (T_w - T_f)$
- $Q_w = 0.75 \times (5.7 \times 10^3) \times (22.2 - T_f)$
- $Q_w = 94905 - 4275T_f$ [1 mark]

Calculate the final temperature of the water, T_f:

- Energy gained by ice = energy lost by water
- $54681.6 + 684T_f = 94905 - 4275T_f$
- $4959T_f = 40223.4$
- $T_f = 8.11$ °C [1 mark]

[Total: 5 marks]

Energy gained by ice = energy required to heat ice + energy required to melt ice + energy required to heat water

because "energy required to heat water" is the energy lost by the ice on the water

1c

(c) Determine through calculation whether the final temperature would be obtained quicker when using the cubes instead of the spheres:

List the known quantities:

- Volume of one sphere = 3.35×10^{-5} m^3 (from part (b))
- Radius of sphere, $r = 20 \times 10^{-3}$ m (from part (b))

Explanation:

- The greater the surface area to volume ratio of the ice pieces then the quicker thermal transfer will take place; [1 mark]

Calculate the surface area to volume ratio of the spheres:

- Surface area of one sphere = $4\pi r^2 = 4\pi \times (20 \times 10^{-3})^2 = 5.03 \times 10^{-3}$ m^2
- $\dfrac{surface\ area}{volume} = \dfrac{5.03 \times 10^{-3}}{3.35 \times 10^{-5}} = 150.15$ m^{-1} [1 mark]

Calculate the side length of the cubes:

- Volume of sphere = volume of cube
- Volume of cube = l^3
- So, $l = \sqrt[3]{volume\ of\ sphere} = \sqrt[3]{(3.35 \times 10^{-5})}$
- $l = 0.032$ m [1 mark]

- Surface area of one cube = $6 \times (0.032)^2 = 6.14 \times 10^{-3}$ m^2
- $\frac{surface\ area}{volume} = \frac{6.14 \times 10^{-3}}{3.35 \times 10^{-5}} = 183.3\ \text{m}^{-1}$ [1 mark]

Conclusion:

- The ice cubes have a higher surface area to volume ratio, so the final temperature would be obtained quicker; [1 mark]

[Total: 5 marks]

2a

(a) Calculate the temperature of the ball immediately before it hits the water:

List the known quantities:

- Height of fall, h = 0.75 m
- Initial velocity of ball, u = 20 m s^{-1}
- Acceleration due to free-fall, g = 9.81 m s^{-2}
- Initial temperature of ball, T_i = –2.1592 °C

Calculate the final velocity of the ball just before it hits the surface of the water, v:

- Recall an equation of uniform motion: $v^2 = u^2 + 2as$
- So, $v = \sqrt{u^2 + 2ah} = \sqrt{20^2 + (2 \times 9.81 \times 0.75)} = 20.36\ \text{m s}^{-1}$ [1 mark]

Consider the energy transfer taking place as the ball falls:

- Initial gravitational potential energy + initial kinetic energy = final gravitational potential energy + final kinetic energy + thermal energy transferred to surroundings
- $mgh + \frac{1}{2}mu^2 = 0 + \frac{1}{2}mv^2 + mc\Delta T$ [1 mark]

Calculate the change in temperature of the ball bearing, ΔT:

- $\Delta T = \frac{2gh + u^2 - v^2}{2c}$

- $\Delta T = \dfrac{(2 \times 9.81 \times 0.75) + 20^2 - 20.36^2}{(2 \times 440)}$
- $\Delta T = 2.1068 \times 10^{-4}$ °C [1 mark]

Calculate the temperature of the ball immediately before it hits the water, T_f:

- $\Delta T = T_f - T_i$
- $T_f = \Delta T + T_i = (2.1068 \times 10^{-4}) + (-2.1592) = -2.1590$ °C [1 mark]

[Total: 4 marks]

2b

(b) Calculate the final temperature of the ball and the water after impact:

List the known quantities:

- Mass of ball bearing, m_b = 33 g = 33 × 10^{-3} kg (from part (a))
- Volume of pool, V = 0.05 × 0.1 × 0.3 = 1.5 × 10^{-3} m^3
- Density of water, ρ = 1030 kg m^{-3}
- Specific heat capacity of water, c_w = 4300 J kg^{-1} °C^{-1}
- Specific heat capacity of ball, c_b = 440 J kg^{-1} °C^{-1} (from part (a))
- Initial temperature of water, T_w = 22.5173 °C
- Temperature of ball upon hitting the water, T_b = –2.1590 °C (from part (a))

Calculate the mass of the water in the pool, m_w:

- Recall the equation for density, $\rho = \dfrac{m}{V}$ so $m_w = \rho V$
- $m_w = 1030 \times (1.5 \times 10^{-3}) = 1.545$ kg [1 mark]

Consider the energy transfer taking place when the ball is in the water:

- Energy lost by ball = energy gained by water
- $m_b c_b (T - T_b) = m_w c_w (T - T_w)$
- $m_b c_b T - m_b c_b T_b = m_w c_w T - m_w c_w T_w$

- $\left(m_b c_b - m_w c_w\right) T = m_b c_b T_b - m_w c_w T_w$
- $T = \dfrac{m_b c_b T_b - m_w c_w T_w}{\left(m_b c_b - m_w c_w\right)}$ [1 mark]

Calculate the final temperature of the ball and the water, *T*:

- $T = \dfrac{((33 \times 10^{-3}) \times 440 \times -2.1590) - (1.545 \times 4300 \times 22.5137)}{((33 \times 10^{-3}) \times 440) - (1.545 \times 4300)}$
- $T = 22.6577$ °C [1 mark]

[Total: 3 marks]

3a

(a) Calculate the overall increase in the temperature of the brake discs:

List the known quantities:

- Mass of cyclist, m_c = 50 kg
- Mass of bike, m_b = 5 kg
- Constant speed, v = 25 m s^{-1}
- Mass of 1 brake disc, m_d = 250 g = 0.25 kg
- Specific heat capacity or brake fluid, c = 1.77 kJ kg^{-1}.= 1.77 × 10^3 J kg^{-1}

Obtain an equation for the conservation of energy as the cyclist comes to a stop:

- $m = m_b + m_c = 55$ kg
- Kinetic energy lost by cyclist and bike = thermal energy gained by 4 brake discs
- $\frac{1}{2} m v^2 = 4\, m_d\, c \Delta T$ [1 mark]

Calculate the increase in temperature of the brake discs, Δ*T*:

- $\Delta T = \dfrac{m v^2}{8 m_d c}$
- $\Delta T = \dfrac{55 \times 25^2}{8 \times 0.25 \times (1.77 \times 10^3)}$
- $\Delta T = 9.71$ °C [1 mark]

[Total: 2 marks]

3b

(b) The temperature difference in the brake disc fluid can be reduced by:

Any **two** from:

- Reduce the mass of the bike and the cyclist; [1 mark]
- Use a fluid with a lower specific heat capacity; [1 mark]
- Increase the mass of the brake discs; [1 mark]
- Increase the number of brake discs used on the bike; [1 mark]
- Increase the surface area of the brake discs; [1 mark]

[Total: 2 marks]

Use the **equation** in **part (a)** and identify the quantities that can be reduced / increased the make the temperature change smaller.

3c

(c) Explain whether adding water to the mineral oil in the brake fluid will make it safer when it overheats:

List the known quantities:

- Specific heat capacity of mineral oil = 1.77×10^3 J kg^{-1} (from part (a))
- Boiling point of mineral oil = 300 °C
- Specific heat capacity of water = 4200 J kg^{-1} °C^{-1}
- Boiing point of water = 100 °C

Explain the effect of adding water on the specific heat capacity:

- Water has a higher specific heat capacity than mineral oil
- So, adding water will mean that the brake fluid requires more energy to increase its temperature; [1 mark]

Explain the effect of adding water on the boiling temperature:

- Water has a lower boiling temperature than mineral oil
- So, when the water in the mixture boils the mixture will compress and the brakes

- Increase the mass of the brake discs; [1 mark]
- Increase the number of brake discs used on the bike; [1 mark]
- Increase the surface area of the brake discs; [1 mark]

[Total: 2 marks]

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- Water has a higher specific heat capacity than mineral oil
- So, adding water will mean that the brake fluid requires more energy to increase its temperature; [1 mark]

Explain the effect of adding water on the boiling temperature:

- Water has a lower boiling temperature than mineral oil
- So, when the water in the mixture boils the mixture will compress and the brakes will not work; [1 mark]

Explain whether adding the water makes using the brakes safer when they overheat:

- Adding the water means that the fluid will heat up slower, some of the fluid will boil at a lower temperature, so the water should not be added; [1 mark]

[Total: 3 marks]